次世代IX研究会におけるルータ相互接続試験

A Report of Router Interoperability from Distix

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Agenda

- Next Generation IX Consortium
  - MPLS-IX Architecture
  - Working Groups
- Router Interoperability Test
  - Policy
  - Test Specification
  - Results
- Discussion

Next Generation IX Consortium

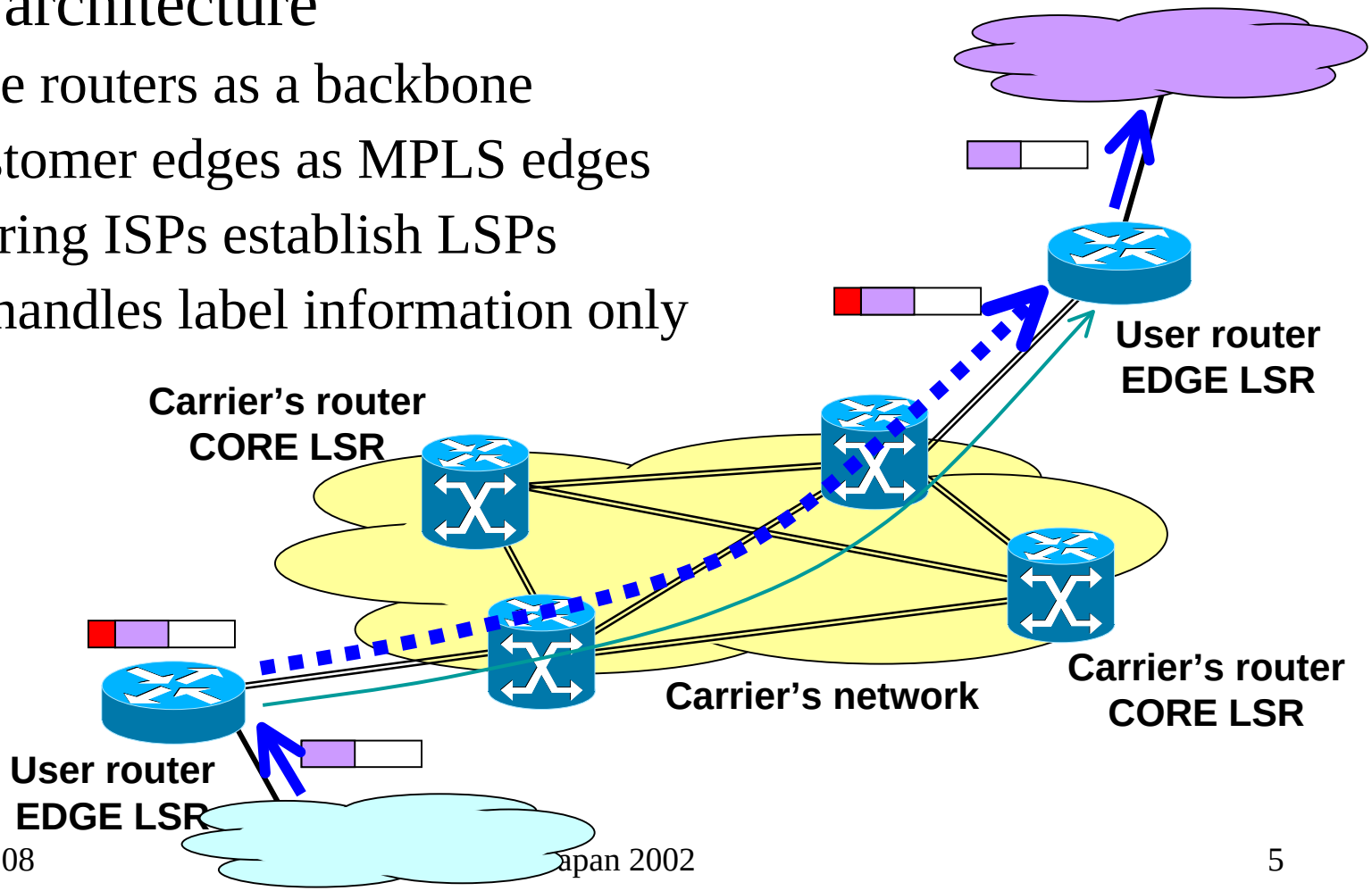
- Background
  - ATM and LAN Equipments has been used for IX
  - Limitation of Scalability (Bandwidth and Flexibility)
- Goal
  - Propose a new IX architecture
  - Verification and Validation on an actual network
  - Sharing know-how about the new technology
- We often call it Distix, but what is Distix?
  - An Ideology? A model? A technology? An test bed?
  - Nobody knows about it correctly. (discovered last night)

Our proposal

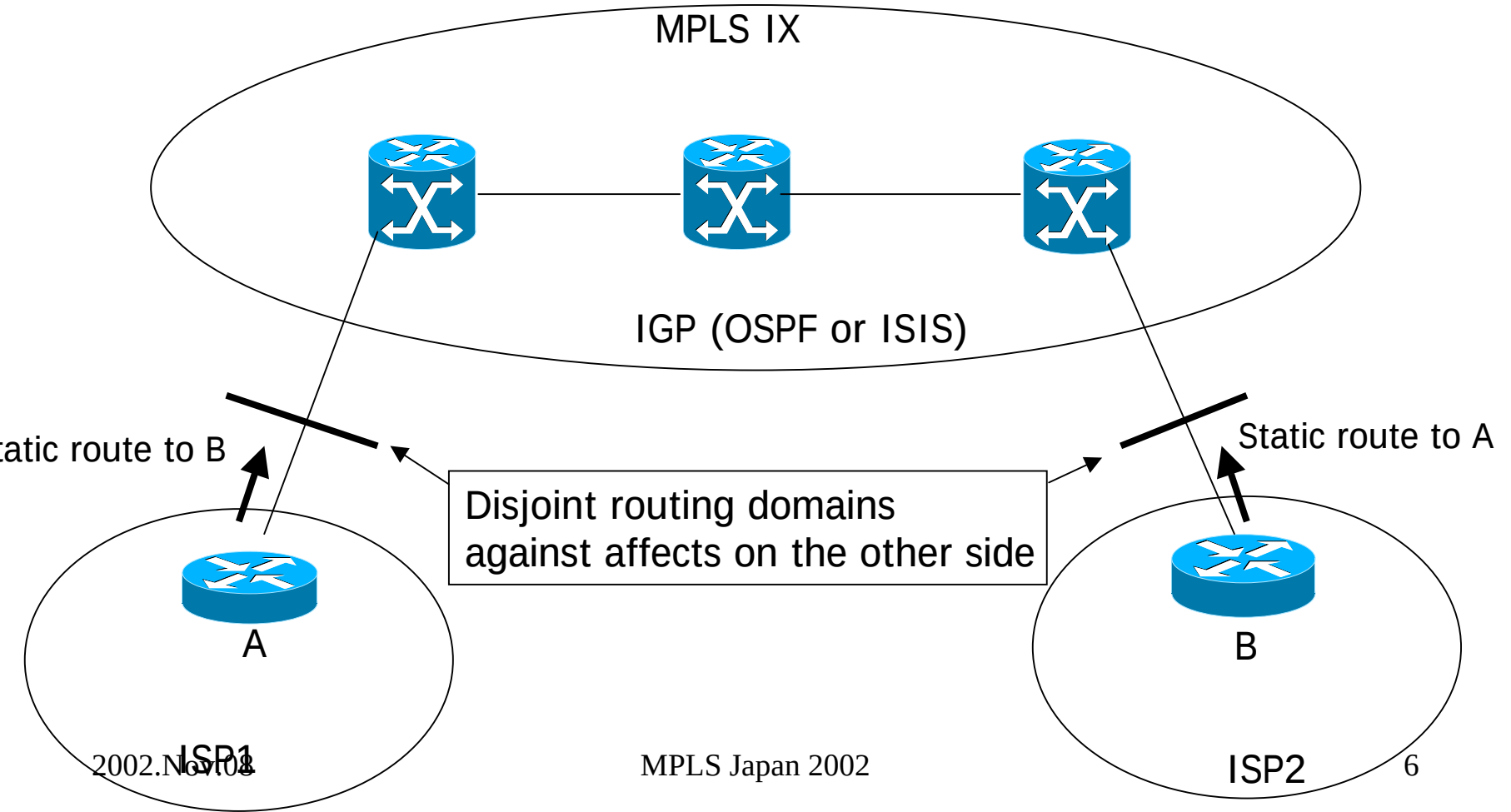
- Requirement of IX
  - Independency from data-link architecture
  - Widely distribution and scalability
  - Easy to maintain
- Let’s use MPLS for IX
  - Allows any L2 arch. (e.g. PoS, GbE, ATM)
  - Easy to extend equipments and exchange points
  - Behaves as one big IX machine

MPLS-IX Architecture

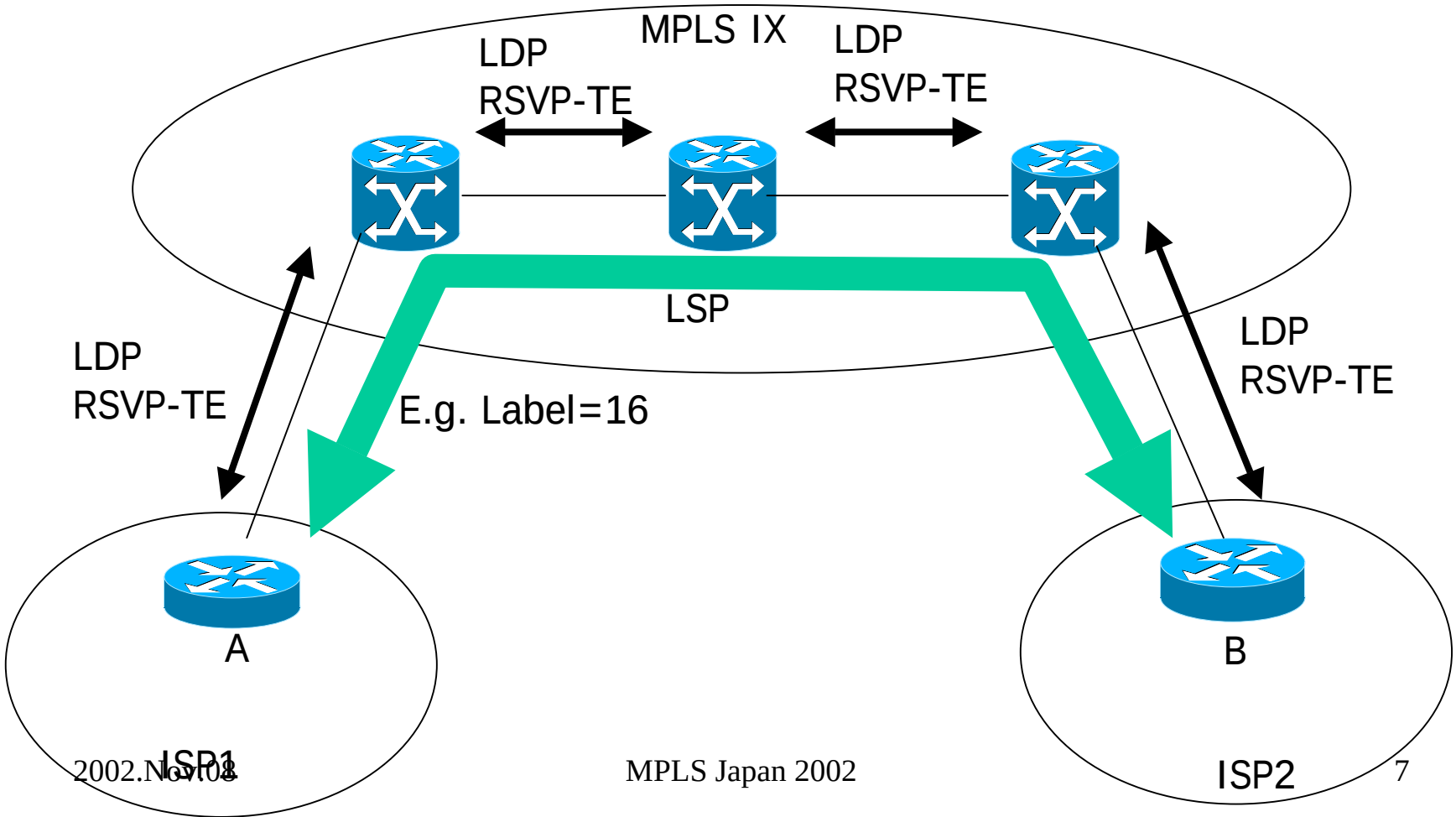
- Basic architecture
  - Core routers as a backbone
  - Customer edges as MPLS edges
  - Peering ISPs establish LSPs
  - IX handles label information only



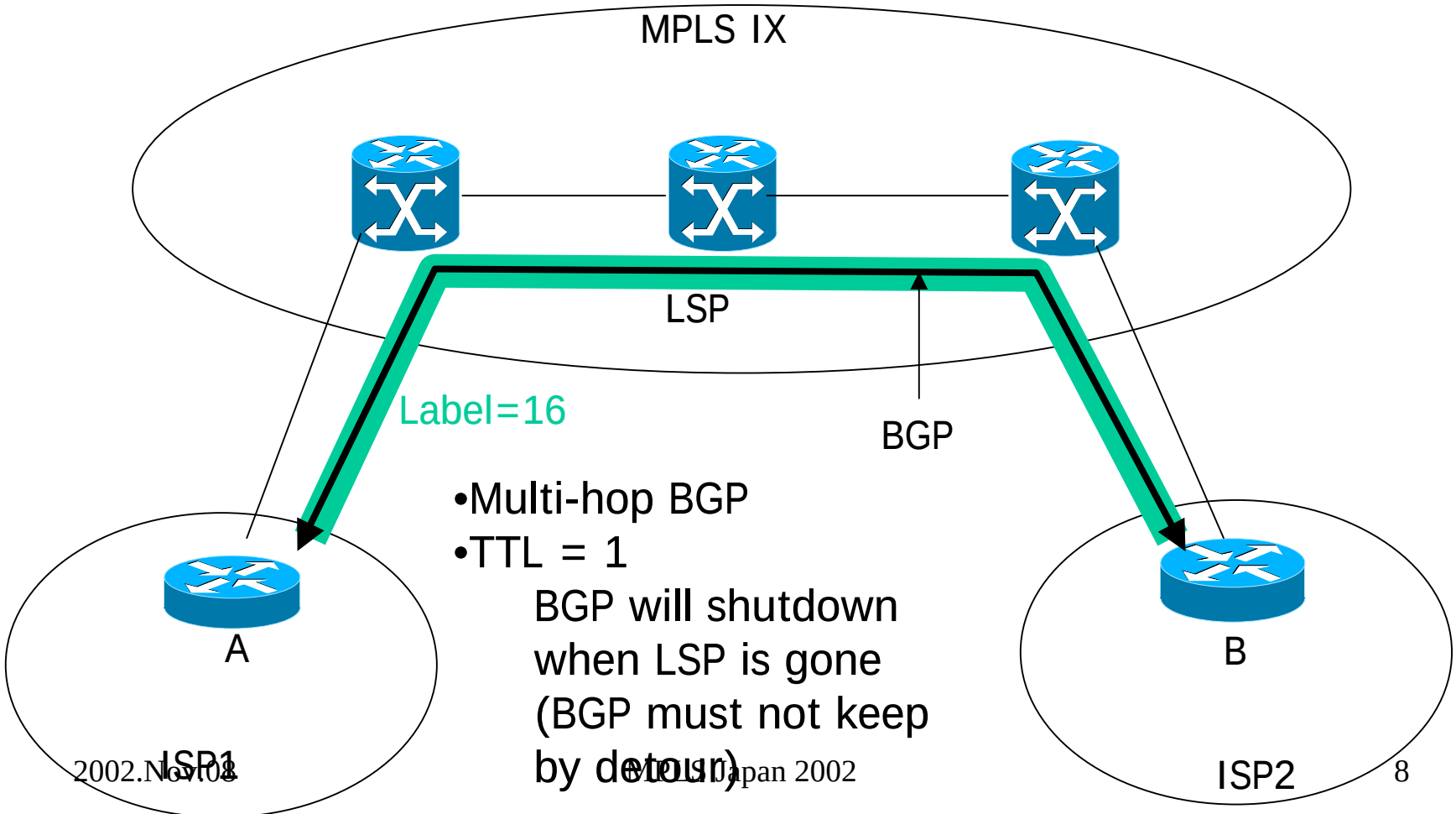
MPLS-IX Architecture (routing domain)



MPLS-IX Architecture (Establishing LSP)



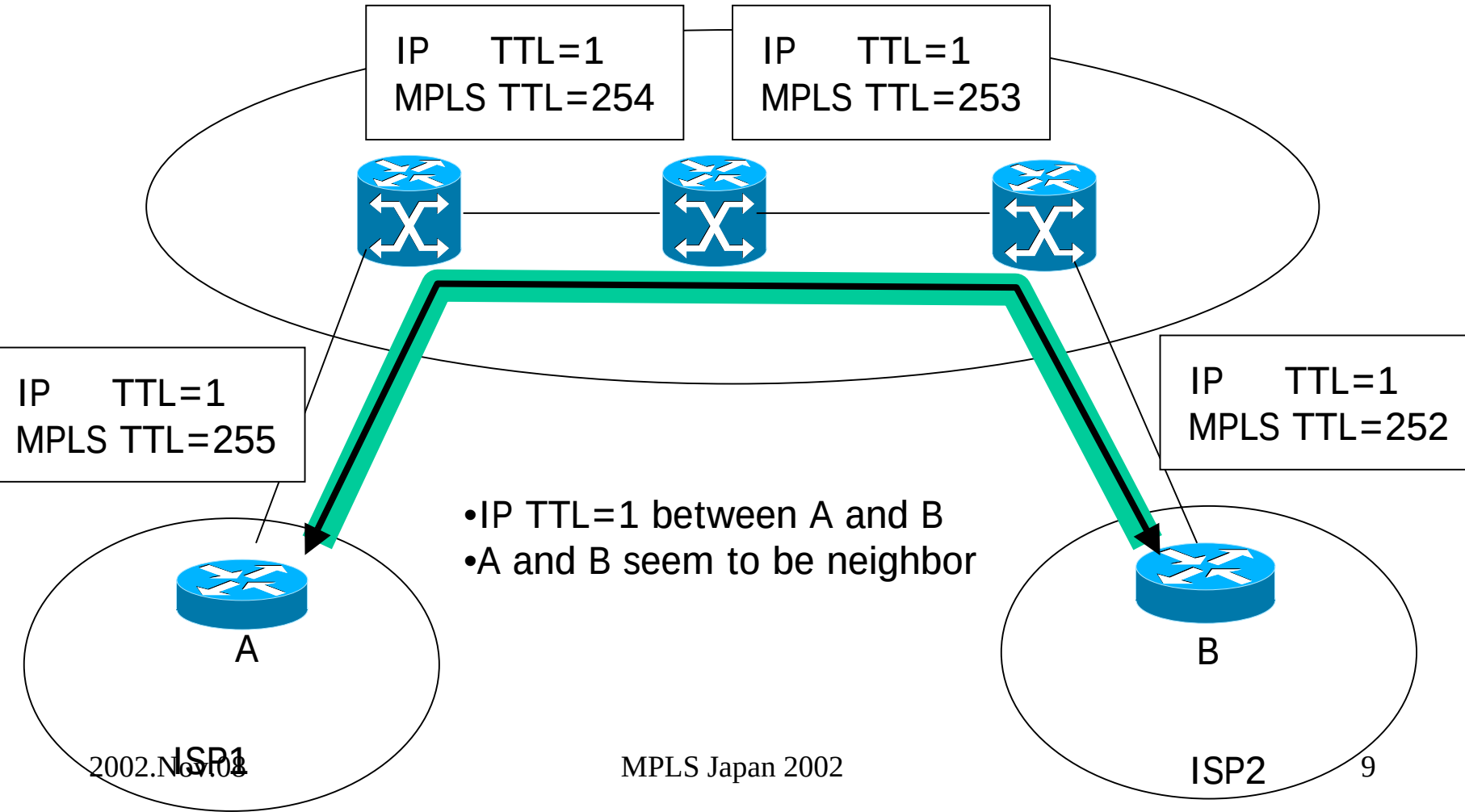
MPLS-IX Architecture (Establishing BGP)



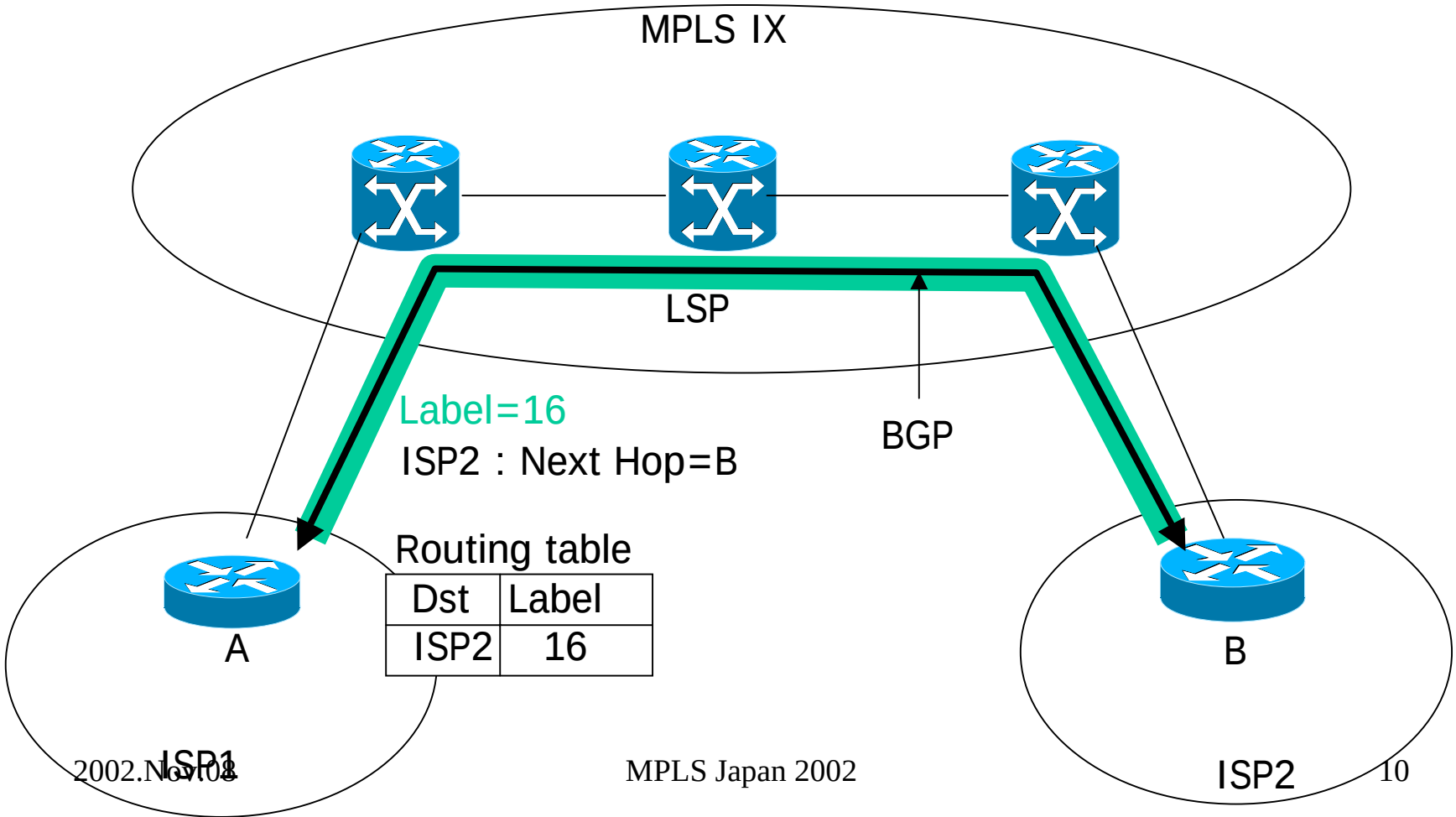
- Multi-hop BGP
- TTL = 1

  BGP will shutdown when LSP is gone (BGP must not keep by detour)

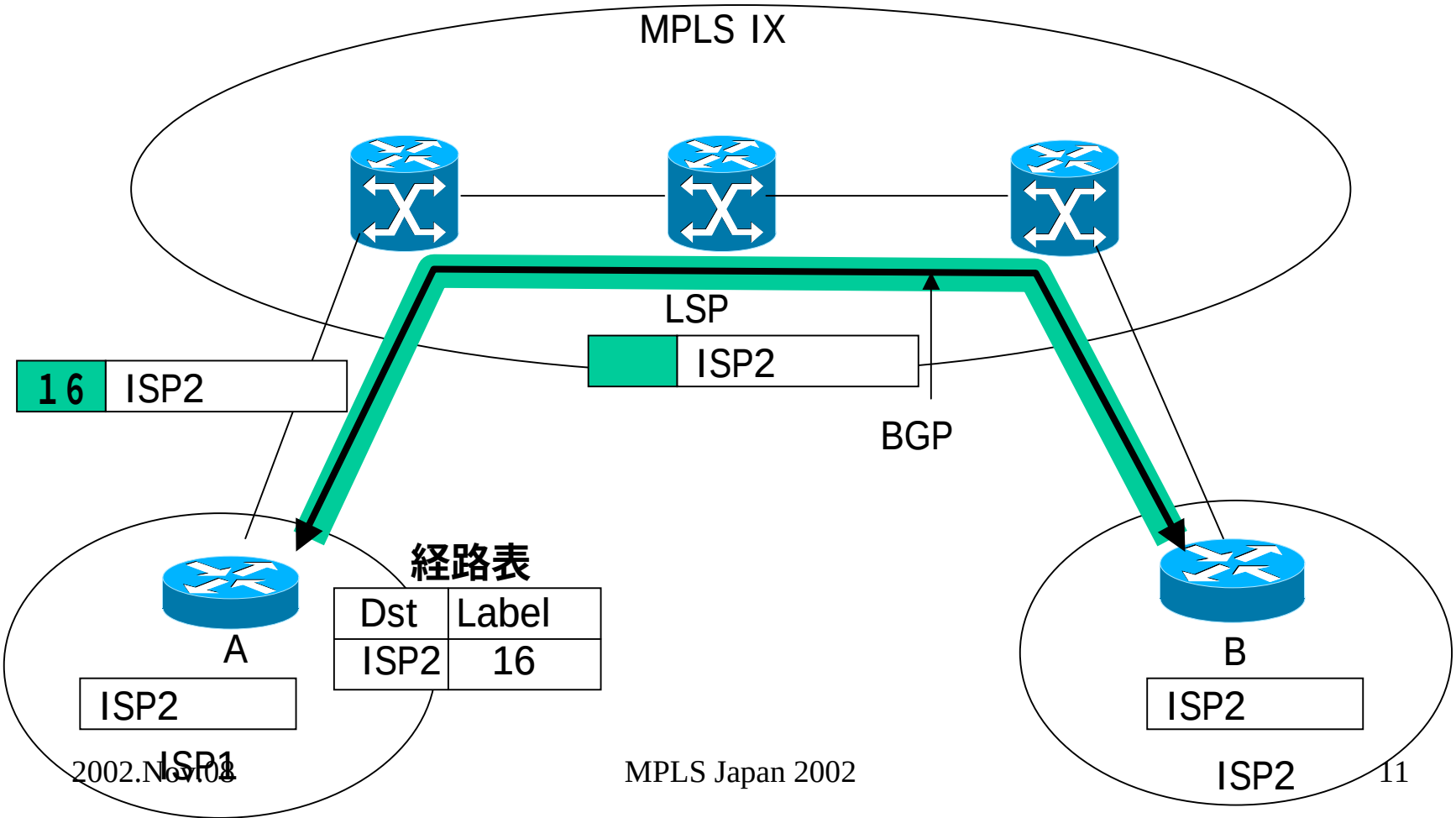
MPLS-IX Architecture (TTL handling)



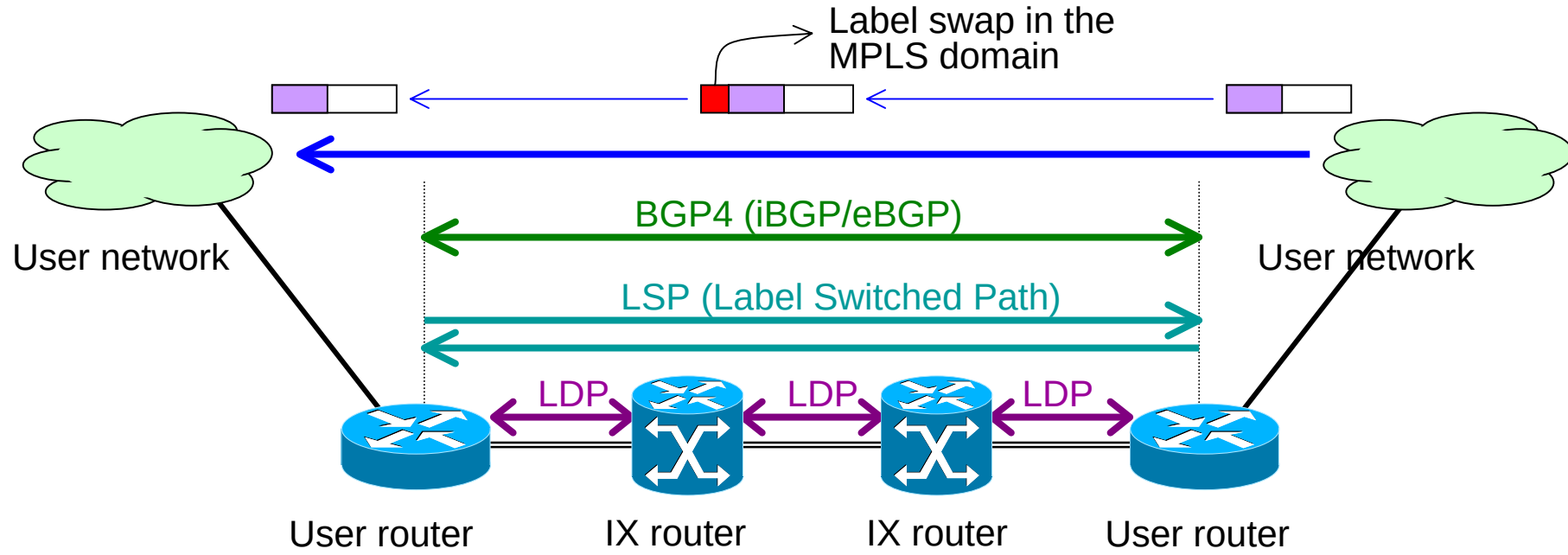
MPLS-IX Architecture (Exchanging routing information)



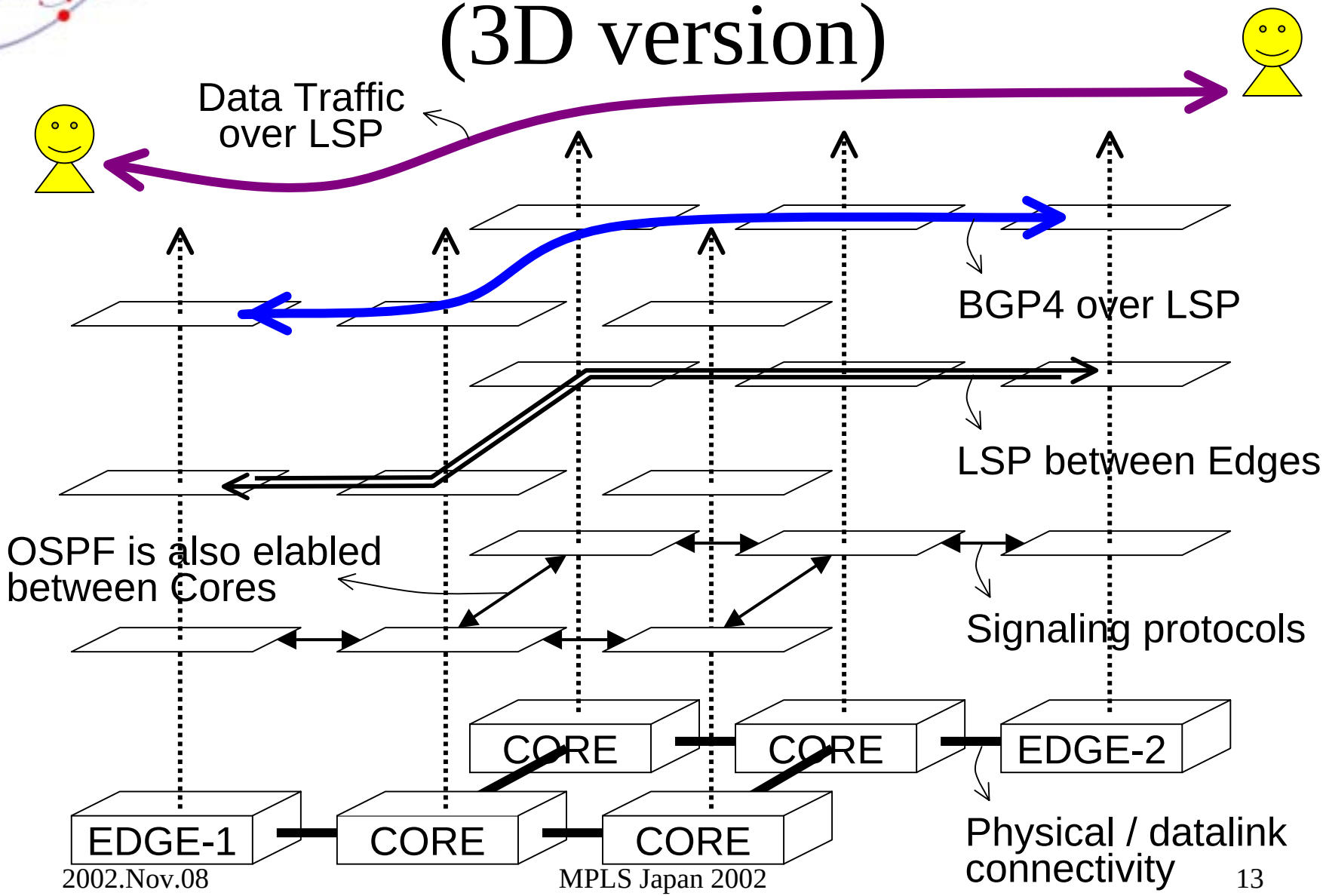
MPLS-IX Architecture (IP datagram forwarding)



MPLS-IX Protocol Design (summary)



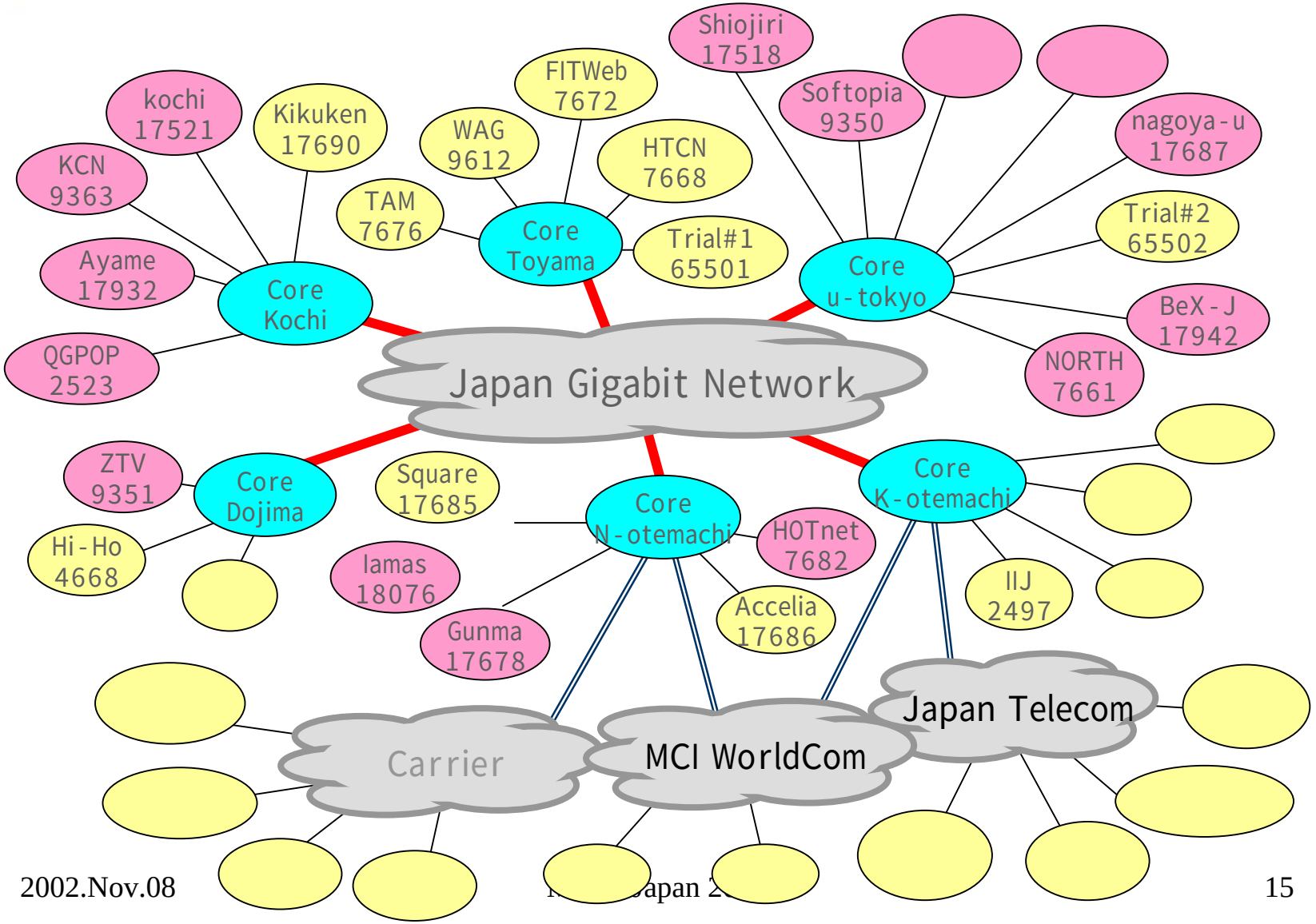
MPLS-IX Protocol Design (3D version)



Working Groups

- IX Users WG
  - Verify the technology by using an actual IX
  - Discuss issues in IX users point of view
- IX Providers WG
  - Verify the technology by operating an actual IX
  - Discuss issues in IX providers point of view
- Router Interoperability WG
  - Verify the interoperability on multi-venders

An Experimental Test Bed



Router Interoperability Test

- Regular Interoperability of MPLS
  - heterogenous router vendors
- Special Interoperability for MPLS-IX
  - Basic Functions
    - Connectivity (LDP/RSVP-TE)
    - Routing (OSPF, BGP4)
    - TTL, Filtering, SNMP support
  - Extended Functions
    - 6PE, RFC3107, LDP over RSVP-TE
    - Redundancy, QoS
- Performance (is not related to interoperability but)

Policy of the Interoperability Test

- Open
  - Participation under the Agreement of Distix
  - Publication the results by WWW
- At Participants own Expense
  - Equipments, Transportation, Stay, and so on
    - Even the site is far from Tokyo
  - No public budget for it
- Help each other, share our knowledge and
- Enjoy!

The Test Steps

- Preparation
  - Specify testing methods
  - Finding the site (difficult to find)
- Test
  - Monday to Friday in a week
  - 0900 to 1900 everyday
- Summarizing the results
  - Classify reliable data
- Publication on the WWW

Required Site Environments

- 100 m^2 at least (10m^2 each per vender)
  - Human/Machine separation are welcome
- AC100V 100A at least
  - AC200V and DC-48V are welcome
- Reachability of Internet and portable phones
- Public transportation
- Accommodation if outside of Tokyo area

History

- 2001.10.15-19
  - TAO JGN IPv6 Center (Tokyo)
- 2002.01.28-02.01
  - TAO Research Center (Chiba)
- 2002.05.13-17
  - Netmarks Quality Management Center (Tokyo)
- 2002.07.01-05
  - N+I 2002 showcase (Chiba)
- 2002.09.02-06
  - TAO Hokuriku IT Center (Ishikawa)

Participants

	exp1	exp2	exp3	N+I	exp4
(Agilent)	○	○	○	○	○
AYAME Project	○	○	○	○	○
Cisco Systems	○	○	○	○	○
Digital Magic Labs.	○	○			
Extreme Networks		○		○	○
Foundry Networks	○	○	○	○	○
Fujitsu	○	○	○	○	○
Furukawa Electric	○	○	○	○	○
HITACHI	○	○	○	○	
Juniper Networks	○	○	○	○	○
NEC (CX)			○	○	○
NEC (IX)	○	○	○	○	○
Netmarks		○	○	○	
(TOYO Technica)				○	
Riverstone Networks	○	○	○	○	○

Test Items

- Edge LSR (as CE)
  - Capability itself, Interoperability
- Core LSR (for IX)
  - Capability itself, Interoperability
- Extended Menu
  - Behavior check in a redundant topology
  - 6PE (IPv6 forwarding with IPv4 control plane)
  - RFC3107 (label carry by BGP)
  - LDP over RSVP-TE
  - Performance (label push, swap and pop)

Test Specification of Edge Capability (General)

- NOT copy TTLs
  - IP TTL to MPLS TTL at Ingress
  - MPLS TTL to IP TTL at Egress
    - If capable of non-PHP
- BGP
  - multi-hop with IP TTL=1
  - BGP packets over LSP
- Check packets really labeled by packet dump
- Check IP forwarding by injecting packets from the out side of Edge LSRs

Test Specification of Edge Capability (LDP)

- Advertisement labels (≧ 16) without PHP
- LSP Filter (according to FEC by signaling)
- Transport address of LDP session
  - Loopback address and/or interface address
- Advertisement FEC
  - Loopback address and/or interface address
- SNMP capability
  - Traffic of LSP
  - Trap of LSP up/down
  - Trap of LDP session up/down

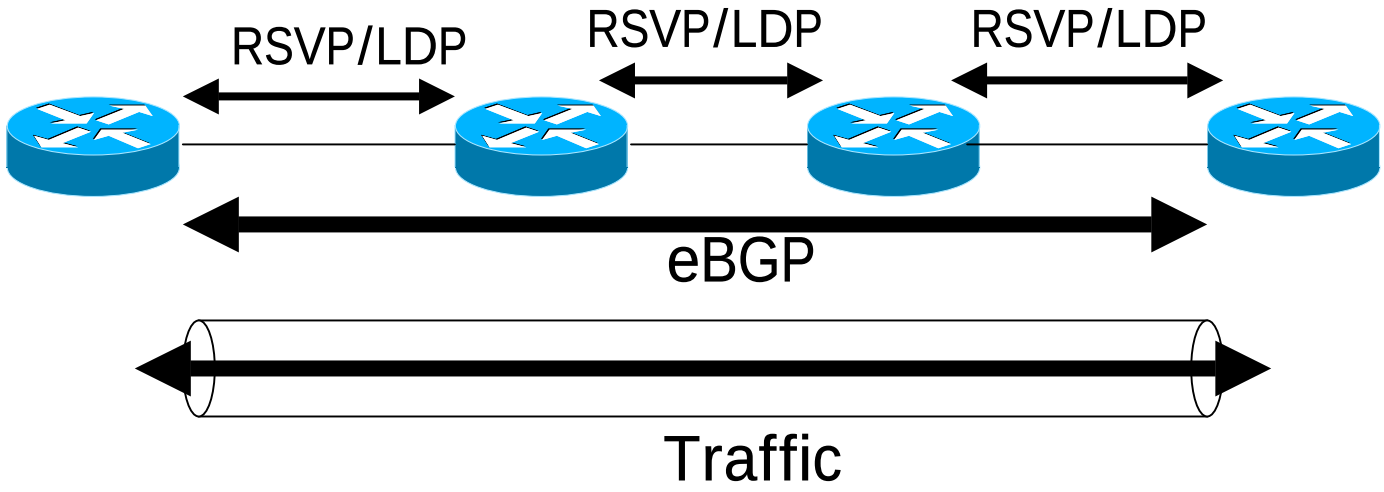
Test Specification of Edge Capability (RSVP-TE)

- Advertisement labels (≧ 16) without PHP
- LSP Filter (according to FEC by signaling)
- Src/dst of RSVP PATH message
  - Loopback address and/or interface address
- SNMP capability
  - Traffic of LSP
  - Trap of LSP up/down
  - Trap of LDP session up/down

Test Specification of Edge Interoperability (common)

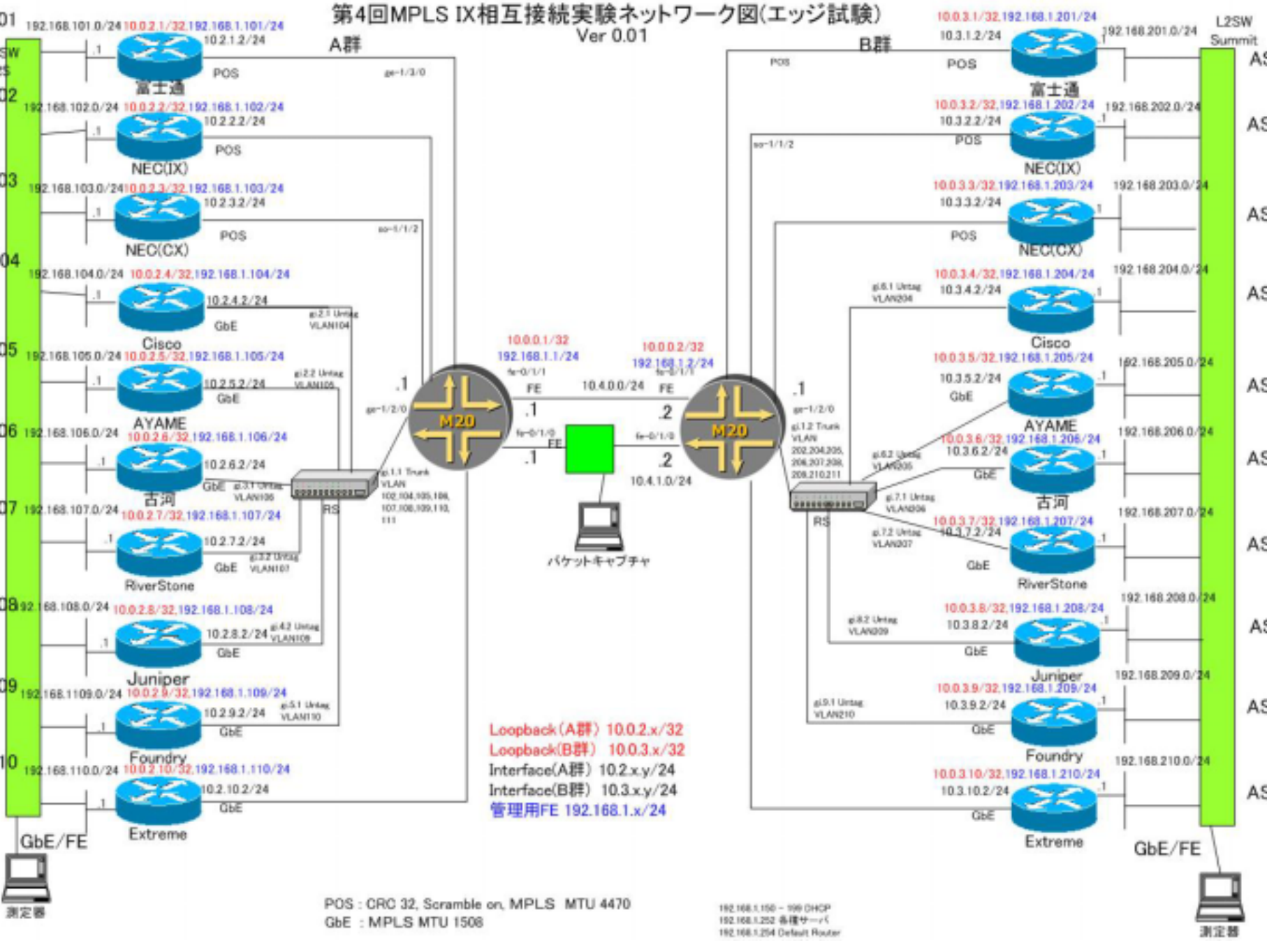
- Establish an LSP
- Exchange BGP packets over the LSP
- Exchange routing information
  - by BGP multi-hop with IP TTL=1
- Not fragment any packets within 1500 bytes
- IP TTL=n+2 at Egress if n at Ingress

Edge Interoperability Test



第4回MPLS IX相互接続実験ネットワーク図(エッジ試験)

Ver 0.01



Ishihara's Ping Engine

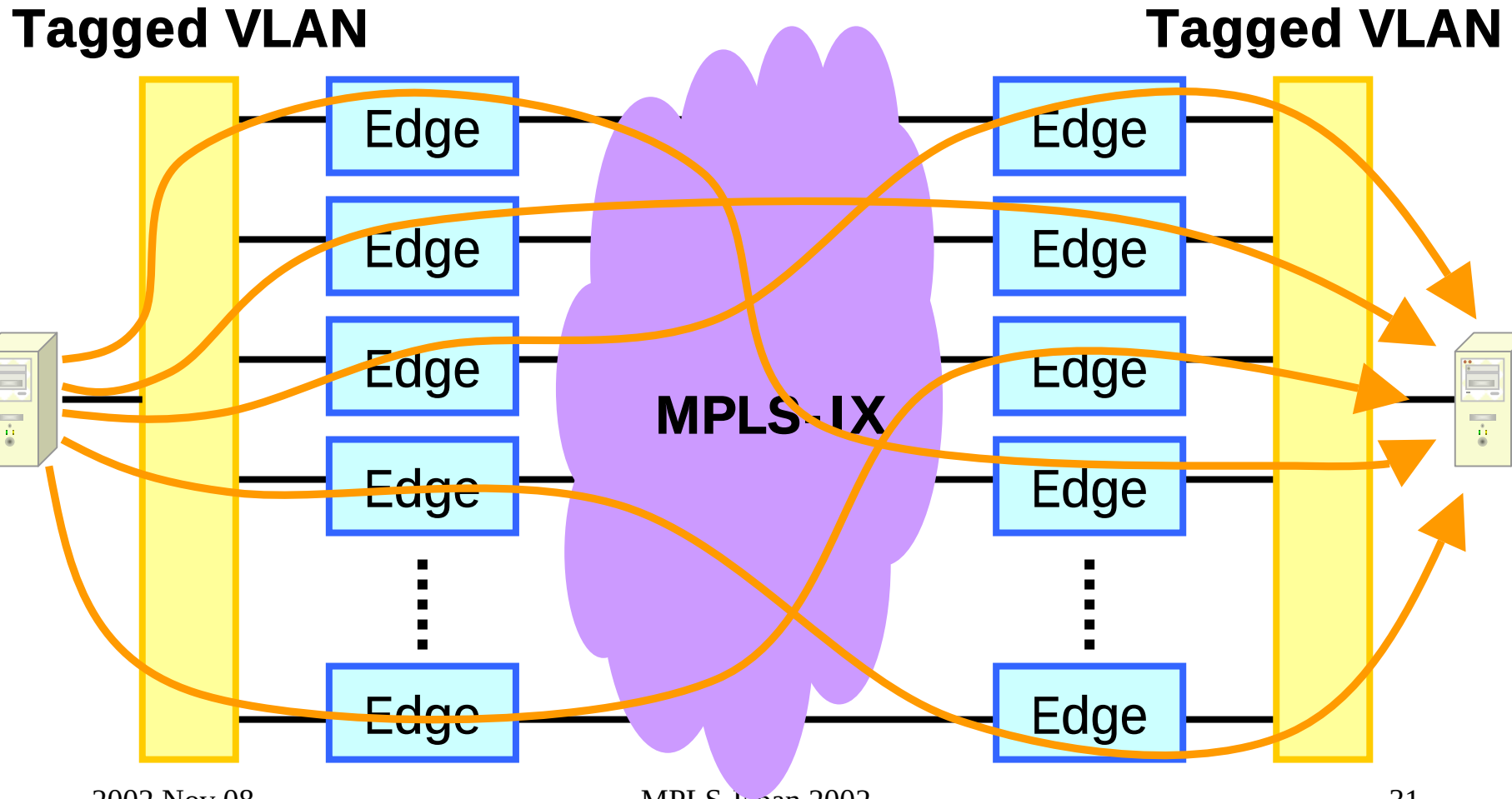
- UDP one-way ping
  - Send a UDP datagram (specified length)
  - Check by the reception of the datagram
  - Tagged VLAN capability
- Implementation
  - Wild lab. LAMB-RT-01SP/32
  - FreeBSD 4.6-Release
  - Software constructed on hping and libnet
    - http://www.hping.org/
    - http://www.packetfactory.net/libnet/

Views of the Ping Engine





Ping Engine with tagged VLANs



Test Specification of Core Capability (General)

- NOT copy MPLS TTL to IP TTL
  - Only if capable of PHP
- Advertisement interface addresses of edge routers (/32) via OSPF

Test Specification of Core Capability (LDP/RSVP-TE)

- Filter for receiving LSPs
- SNMP capability
  - Traffic of LSP
  - Trap of LSP up/down

Test Specification of Core Interoperability (Common)

- Check at the edge LSRs
  - Establish an LSP
  - Not fragment any packets within 1500 bytes
  - IP TTL=n+2 at Egress if n at Ingress

Sample Results

- Edge capability
  - General, LDP, RSVP-TE
- Edge interoperability
  - LDP (loopback, Interface)
  - RSVP-TE (loopback)
- Snapshots by Ishihara's ping engine
  - RSVP-TE (interface)
    - Short packets, Long packets

Testing View (1)



Testing View (2)




networks

Testing View (4)



Testing View (5)

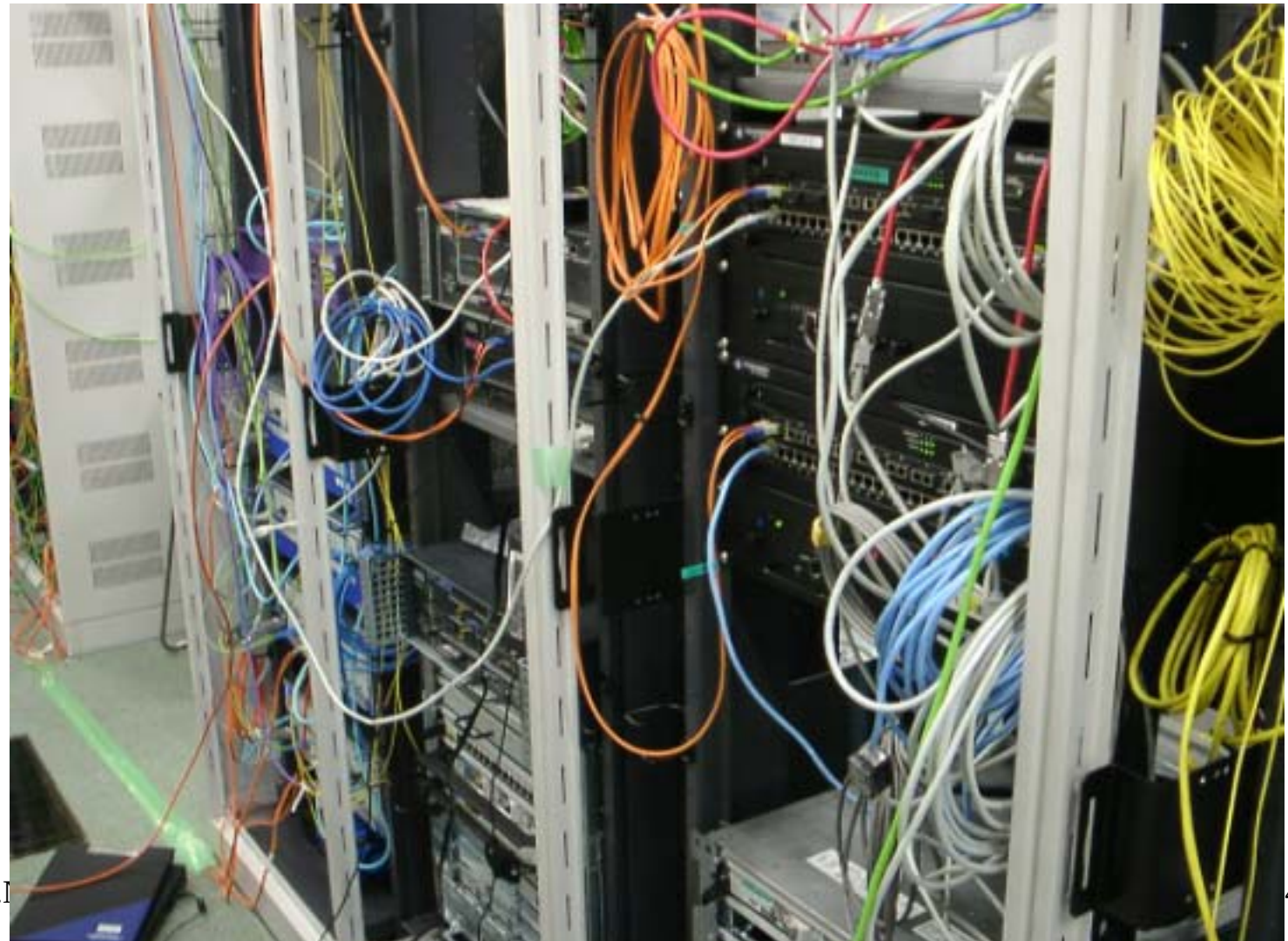



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Testing View (7)



Testing View (8)



Testing View (9)



Summary of the Past Results

- Basically sufficient
  - Still some troubles may occur by design, implementation, and versions.
  - Does mostly interoperability depend on what signaling they implement, LDP v.s. RSVP-TE?
- Insufficient in an operational point of view
  - Without PHP
    - IX Users have a thirst for labels at egress LSRs
  - SNMP support
    - MPLS-IX consists of many routers even at a distance

Discussion

- Verification of the interoperability for MPLS-IX
- Reconfirmation requirements and design of the MPLS-IX architecture
- Contribution to MPLS community
  - Attractive and Important Experiences
- A wall of Complexity of $O(n^2)$
- Shift to new Features
- A new direction will be needed
  - We will pass a sub goal in the next (5th) test
  - Collaboration with other WGs is necessary

See in detail

- Distix Home Page
  - http://www.distix.net/
- Distix Router Interoperability WG
  - http://www.distix.net/router-wg/
- 5th Interoperability Test in Kochi
  - Coming up soon!
  - Venders,
    - Do not forget sending an application of participation

Thank you!

- To all the participants of the interoperability tests
- To the audience
- Distix was sponsored by TAO I121-401

